A hybrid deep learning based robust framework for cattle identification

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2022 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC) | 978-1-6654-6109-2/22/$31.00 ©2022 IEEE | DOI: 10.1109/ASSIC55218.2022.10088414

***Abstract*—This study proposes a deep learning-based framework for recognizing cows based on images of their muzzle, and faces. This method works well when dealing with missing or false insurance claims. This study proposes a hybrid multi-stage framework consisting of different phases such as augmentation, denoising, enhancement, and classification. The proposed framework is developed by hybridizing convolutional denoising autoencoders (CDAE), least squares generative adversarial network (LS-GAN), Xception feature extractor, and a convolutional neural network (CNN). CDAE is used to initiate the process of denoising noisy images. LS-GAN is used to improve the characteristics of denoised images by enhancing the image by elimination of the residual noise. The Xception is utilised to extract significant and optimal features, and CNN is then used for classification. Various comparative methodologies are used to assess the proposed approach at different phases through several statistical measures. The proposed framework achieved 97.27% accuracy using the test datasets, which is higher than the comparative approaches.**

***Index Terms*—cow face features, muzzle, GAN, autoencoder, hybrid method**

I. Introduction

A. Background

Animal identification refers to the process of correctly identifying cattle in the agricultural and animal husbandry industries. Its applications in precision farming might include measuring weights, welfare evaluation and fitness tracking. Cattle identification benefits stakeholders, farmers, livestock producers, the food sector, and so on. Because of animal identification, livestock immunisation regimens may be arranged more effectively, reducing mismatches. Furthermore, animal identification aids in the determination of livestock losses and facilitates livestock control in large-scale farm regions [1]. RFID, tattooing, tagging, and ear notching are some of the classic cattle identifying systems. Shape deformations, tag losses, fraud, animal welfare issues, and restricted scalability are some of the problems [2]. The modern world need unique cattle identification systems that make use of cutting-edge technology and research. Visible livestock biometrics has emerged as an important topic in computer vision, and it may be the most effective method of recognising cattle.

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B. Previous works

Authors used three sorts of techniques in this sector, according to studies and research: naive method, machine learning strategy, and deep learning strategy. With 89.95% accuracy, the naive technique used visual recognition by branding, ear notching, tattooing, tagging, radio frequency identification (RFID), or nose printing [3] and visual oriented cow recognition using body patterns such as patches and spots [4]. Machine learning techniques used to animal biometrics to identify individual farm animals are a potential option that addresses many of the issues associated with traditional identification [5]. The ensemble approaches such as weighted voting, support vector machine (SVM), and random forest are also employed The authors proposed Shi-Tomasi corner for the cattle identification [6]. SVM achieved 90% accuracy [7]. In the deep learning, the recurrent neural networks such as long short-term memory (LSTM) and bidirectional long short-term memory (Bi-LSTM) are also used. In the [8], the accuracy was 88%, and 91% for the 15, and 20-frame video duration on 41 farm animals [7]. The parameter transfer method is used in the [9], [10]. Some authors also used pre-trained models such as VGG16, and ResNet50 for transfer learning approaches [11]. The hybrid approaches are also developed for the improvement of the accuracy. For example, the authors presented hybrid frameworks for farm animal identification based on SIFT and RANSAC algorithms [12]. The removal of redundant noise, on the other hand, is critical. Despite the fact that many authors have provided alternative methodologies for feature preprocessing and extraction, it has been shown that ordinary CNN no longer produces the required results with bigger and more complicated datasets. The majority of methods were complex with high computational costs.

C. Contributions

Hence, the fundamental goal of this study is to develop a hybrid multi-phase framework for the cattle identification using face and muzzle images. The following are the study's major contributions:

a) There is no widely available resource of cow face, and muzzle image database. Face, and muzzle image database has been prepared.

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b) CDAE is used to eliminate the noise present in the raw image database.
c) LS-GAN is utilized to enhance the denoised images by eliminating the residual noise present.
d) Extraction of a textural features from cattle face, and muzzle images and classification using CNN.
e) The proposed framework outperforms benchmark techniques for cow identification using muzzle, and face images characteristics, according to experimental study.

D. Organization

The rest of the paper is organized as follows: Section II demonstrates the proposed methodological principles. Section III presents the experimental results and validation of the proposed framework. The conclusions are shown in the Section IV, and the references follow.

II. Proposed Methodology

The architecture of the proposed framework is demonstrated in this section. The proposed framework is a generic framework useful for solving any cattle classification challenge. There are five phases present: preprocessing, denoising, enhancement, extracting features, and classification as shown in the Fig. 1. Each phase is described in detail in the subsections that follow.

A. Data Pre-processing

The collected dataset includes cow-face, and muzzle images from 20 distinct types of cows having 20 (face:10, and muzzle:10) distinct images for each cow. Thus, making the size of dataset equals to 400 images. There are various factors that might make acquiring adequate data difficult, if not impossible. This constraint can be solved using the image augmentation technique [13]. At this phase, additional images are created by employing 32 different augmentations, hence producing a larger face, and muzzle image collection from the existing dataset. Each cow class now has 640 images, for a total dataset size of 12800, i.e. (32 augmentations * 20 images * 20 cows) image dataset.

B. Image Denoising using deep convolutional DAE

The prepared dataset contains a large number of noisy images. These noisy images impacts the performance of the classification task. Hence, this study utilized a CDAE network to eliminate the noise present in the raw images. The common challenges present in the conventional autoencoders like ambiguity and overfitting, can be solved by CDAE. Encoder and decoder are the two parts present in the CDAE. The encoder's goal is to produce the latent form from the input image consisting of the dominant characteristics with lower uncertainty. From the latent form, the decoder extracts the characteristics of the input for denoising the images. CDAE, like the conventional autoencoder, is a feed forward network that is trained by applying the estimated gradient reduction of the inverse log-likelihood loss function (LF) as shown below.

$$LF = -\log(p_{CDAE}(x|\hat{x})) \tag{1}$$

The noisy images obtained after phase 1 are denoised using the CDAE. However, the residual noises are still present in these images which hinders the feature extraction phase. Thus, the image enhancement is performed using LS-GAN as demonstrated in the next subsection.

C. Image Enhancement with LS-GAN

The residual noises left over after image denoising with CDAE are critical for the feature extraction stage. Hence, for eliminating the residual noise and enhancing the images for the better feature extraction, LS-GAN is utilised in this study. The conventional GANs suffers from the problem of vanishing gradients and loss saturation. Hence, in this work LS-GAN is employed [14]. The developed LS-GAN are more stable than the conventional GANs. In the LS-GAN, the adversarial training principles helps in the upscaling and enhancement of the images. The LS-GAN show increased precision and mean opinion ratings (MOR). The LS-GAN is made up of generator and discriminator. The generator uses a number of CNN layers, Batch-normalization layers, and a parametric ReLu for the activation function. The discriminator checks if the image is produced has enhancement. The generator and the discriminator are trained in adversarial method for generating the enhanced images eliminating the residual noise. By penalizing samples that are on the right side of the decision boundary, LS-GANs stimulate the generator to provide more effective samples. As the generator is updated, this penalty also aids in producing new gradients, which stabilizes the training process by preventing the vanishing gradient issue. These enhanced images are fed to the feature extractor for the extraction of the important characteristics from the images. The objective functions of the generator and discriminator of the developed LS-GAN for the image enhancement are given below, where D is the discriminator, and G is the generator, x is the samples generated, z is the input to the generator.

$$\min_{D} V_{(D)} = \frac{1}{2}\mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}(\boldsymbol{x})}\left[(D(\boldsymbol{x})-1)^2\right] + \frac{1}{2}\mathbb{E}_{\boldsymbol{z}\sim p_{\boldsymbol{z}}(\boldsymbol{z})}\left[(D(G(\boldsymbol{z})))^2\right]$$

$$\min_{G} V_{(G)} = \frac{1}{2}\mathbb{E}_{\boldsymbol{z}\sim p_{\boldsymbol{z}}(\boldsymbol{z})}\left[(D(G(\boldsymbol{z}))-1)^2\right].$$

D. Feature Extraction

In the classification, feature extraction is critical for improving performance. Because natural and raw data are unreliable for direct image classification, optimal and dominant features should be supplied to the classifier for the final prediction. Xception model, is employed in this study to produce optimal features. Xception is a deep learning network made up of 71 layers using over one million images [15]. The input layer of the model is made equivalent to 224*224*3, the dimensional result of phase 3. The developed Xception model is a convolutional neural network which is based on depthwise seperable convolutional layers. In this model, the cross channel correlations, and the spatial correlations are decoupled. There are 36 convolutional layers are present for

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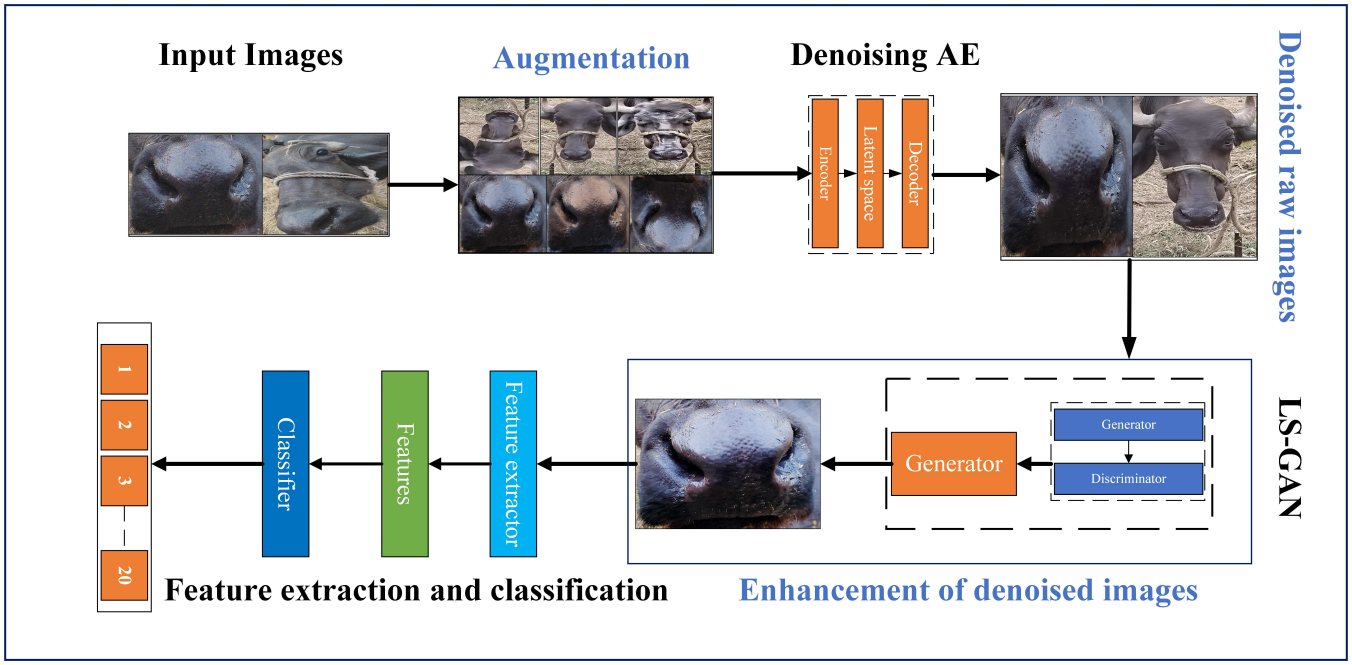

Fig. 1. Representation of the proposed framework for cattle identification using muzzle, and face images

the feature extraction which are structured into 14 modules. After the feature extraction part, the features are flattened using a fully connected layer representing the latent representation of the recovered attributes from the input images.

E. Classification

For each image, the feature extraction step retrieves the optimal and prominent features from the phase 4. The obtained attributes are then transferred to a the 1D CNN network for classification. The 1D CNN layers are utilized with a ReLU function and a dropout layer comprise the classifier network. To prevent overfitting, the dropout layer is used. The output layer consists of 20 neurons with softmax activation functions, each representing a different class of cow and hence capable of differentiating among 20 cow classes.

III. Results and Discussions

This section shows two parts: database information and experimental results. Different models are used for validation of proposed framework at various phases. The CDAE approach is compared to benchmark approaches such as wavelet approaches for the image denoising phase. Simultaneously, the performance of the LS-GAN is compared against methods such as histogram equalisation, Wiener filter, CLAHE, etc for the image enhancement and residual noise elimination. Similarly, for testing classification performance, the proposed feature extractor technique is compared to existing robust approaches.

A. Database information

A cow face, and muzzle image dataset is prepared from a dairy farm located in Telangana state, India, for this study. The data set comprises of 20 cow images collected from various angles that describe the cows. The images consists of both the muzzle, and the face images. The data gathered consists of 400 muzzle, and face images. The collected data is supplemented and preprocessed to create a new dataset of 12800 images. Custom augmentations are performed to retain image information, enhance the quantity of the provided data, and improve training. This new data is separated into train, validation, and test tests, with the train set including 80% of the data and the validation and test sets containing 10% and 10% of the data, respectively.

B. Evaluation Indices

At each phase of the system, multiple performance metrics are employed to assess the proposed hybrid framework. Different performance measures are used throughout the phases as shown in the Table 1. The structural similarity index (SSIM) value lies in $\{0, 1\}$ with 1 indicating a perfect match between the reconstructed image and the original image. Mean square error (MSE) is a comprehensive reference metric, and lower values are desirable. The higher the value of PSNR, the higher the quality of the compressed or rebuilt image. For the classification, parameters such as accuracy, recall, precision, and F1 score are used. The higher these parameters, the better the classification model.

C. Performance of the proposed framework for classification

The proposed framework and reference comparable approaches are validated using test data at each phase, and the results obtained are reported in Tables II-IV, respectively. From the Tables II, III, and IV, it is observed that the the proposed framework model containing the CDAE have the

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TABLE I
THE EVALUATION INDICES UTILIZED FOR VALIDATING THE PROPOSED HYBRID FRAMEWORK

Measure	Formula	Definition
SSIM(x,y)	$\frac{(2\mu_x\mu_y+C_1)+(2\sigma_{xy}+C_2)}{(\mu_x^2+\mu_y^2+C_1)(\sigma_x^2+\sigma_y^2+C_2)}$	μ: average; σ^2: variance; C: constant
MSE	$\frac{1}{ab}\sum_{i=1}^{a}\sum_{j=1}^{b}(x_{ij}-y_{ij})^2$	x_{ij}: pixel of image; y_{ij}: pixel of stego image
		a: rows in image; b: columns in image
PSNR	$\frac{10\log_{10}[\max(\max(x),\max(y))]^2}{\lvert x-y\rvert^2}$	x,y: windows of size NxN
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	TP: true positive
Recall	$\frac{TP}{TP+FN}$	TN: true negative
Precision	$\frac{TP}{TP+FP}$	FP: false positive
F1 score	$\frac{2*Precision*Recall}{Precision+Recall}$	FN: false negative

TABLE II
RESULTS OF PROPOSED FRAMEWORK AND COMPARATIVE MODELS FOR DENOISING RAW IMAGES

Method	PSNR	SSIM	MSE
Total Variation Filter	23.8453	0.3576	1.0578
fastNlMeansDenoisingColored	15.1057	0.2165	2.3722
Wavelet Visushrink	23.2126	0.3171	1.2334
Wavelet Bayeshrink	23.9117	0.4196	1.0370
Proposed framework	**28.9081**	**0.9706**	**0.2317**

TABLE III
RESULTS OF PROPOSED FRAMEWORK AND COMPARATIVE MODELS FOR ENHANCEMENT OF DENOISED IMAGES

Method	SSIM	PSNR	MSE
Histogram Equalisation	0.8211	16.4568	1.4703
CLAHE	0.8492	20.2133	1.1908
Min-Max Stretching	0.9399	48.1318	0.9997
Median Blur	0.9145	35.0002	1.0562
Proposed framework	**0.9951**	**54.7721**	**0.2167**

lowest MSE of 0.0027 and the greatest PSNR and SSIM of 28.9081 and 0.9706, respectively, when compared to the reference methods such as wavelet transformation techniques and total variation filter.

Wavelet Bayeshrink came in second with PSNR, SSIM, and MSE values of 23.9117, 0.4195, and 1.0369. The proposed framework improved the PSNR and SSIM by 4.9964 and 0.5511, respectively, while decreasing the MSE by 1.0342 indicating the domination of the proposed method in the denoising phase. In the enhancement phase, the proposed model with LS-GAN have the lowest MSE of 0.2167 and the greatest PSNR and SSIM of 54.7721 and 0.9951, respectively. Median Blur and the Wiener filter came in second position and the improvement by the proposed framework in PSNR is by 19.77, SSIM by 0.08, and decreased MSE by 1.8395. The proposed framework obtained the best test accuracy of 97.27%, validation accuracy of 97.85%. Similarly the recall and precision are also higher than the other models in the feature extraction and classification task. The Resnet V2 model occupied the second best spot with 96.13%, 96.88% as validation, test accuracies, respectively. The baseline CNN model failed to obtain the desirable metrics in the feature extraction and occupied the last place in the comparison metrics. This poor performance is due to the complex characteristics of the input database. However, the proposed framework managed to dominate in all the phases with the best metrics compared to the other models. From the results, it clearly indicating that the proposed framework is best suited to address cattle identification using face and muzzle images.

IV. COMPARISON OF PROPOSED APPROACH WITH THE LITERATURE

For better validation, the proposed framework is compared with the existing approaches from the literature for the cattle identification. The observations are tabulated in the Table V. The Table V demonstrates the better performance of the proposed framework for the cattle identification.

TABLE V
IMPROVEMENT OF THE PROPOSED FRAMEWORK OVER THE EXISTING LITERATURE APPROACHES

Model	Accuracy (%)	Improvement (%)
[16]	95	2.27
[8]	91	6.27
[9]	91	6.27
[7]	90	7.27
Proposed framework	**97.27**	**-**

V. CONCLUSIONS

Individual cow identification is a critical topic in computer vision-based livestock management. Hence, this research proposes a hybrid cow identification framework using muzzle, and face images. This study proposes a hybrid multi-stage framework consisting of different phases such as augmentation,

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TABLE IV
RESULTS OF PROPOSED FRAMEWORK AND COMPARATIVE MODELS FOR THE CLASSIFICATION

Model	Val A (%)	Train A (%)	Recall (%)	Precision (%)	F1 Score (%)	Test A (%)
VGG16	97.63	89.00	96.00	96.00	96.00	96.09
InceptionResNet V2	96.13	86.07	97.00	97.00	97.00	96.88
NasNet Large	93.42	89.08	95.00	95.00	95.00	95.61
VGG19	96.63	82.16	96.00	96.00	96.00	95.90
ResnetV2	96.06	90.26	97.00	97.00	97.00	96.88
Proposed framework	**91.65**	**98.00**	**98.00**	**98.00**	**97.00**	**97.27**

Val A:validation accuracy; Train A:train accuracy; test A:test accuracy;

denoising, enhancement, and classification. The CDAE is used to denoise the raw images, LS-GAN is used for enhancement, Xception model for the feature extraction, and CNN for the classification. A cow face, and muzzle image dataset of 20 cows is prepared gathered from a dairy farm located in Telangana state, India, for this study. A comprehensive examination of the proposed framework was completed with performance validation, and it was evaluated on a test dataset of 1280 random images of 20 distinct cows, with an accuracy of 97.27 %. Various conventional methods are used as the reference models for evaluating the proposed framework performance in the three phases: denoising, enhancement, classification. From the experimental results, it is apparent that the proposed framework achieved the best results than all the comparative methods involved.

The authors intend to test this proposed framework approach on a bigger and more complicated dataset in the future. In addition, the proposed framework can estimate nutritional and health-related characteristics for cattle, as well as conduct more sophisticated ownership claim determinations.

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